

ZOOMLION

ZS1218ERT

Operation and Safety Manual

Nove 2023 Version B

Foreword

Zoomlion appreciates your choice of our machine for your application. The Operation and Safety Manual must be read and understood in its entirety before operating the machine.

This manual introduces you safety information, significant technical specs, safety operation in detail for working efficiency improving. Keep this manual properly at all times for looking up.

Do not operate the machine if there is any doubt in operation, please consult local service team for troubleshooting. Zoomlion Intelligent Access Machinery Company does not take the consequence of wrong operation.

This manual should be considered a permanent part of your machine and should remain with the machine at all times.

The content is under intellectual property protection, permission is required for a copy or other application.

There might be some tiny differences in details between your machine and the upgraded one due to the continuous improving. For clarification, questions, or additional information regarding any portions of this manual, Contact Zoomlion Access aftersales.

Our company reserves the right to modify this manual as technical improvement without notice.

Thank you for your trust and support for Zoomlion products!

Zoomlion Intelligent Access Machinery Co., Ltd.

Safety Precaution Icons

This manual has the following safety precaution icons:



Failure to comply with the safety precautions listed in this manual could result in personal injury or death.



Failure to comply with the safety precautions listed in this manual could result in potential personal injury or death.



Failure to comply with the safety precautions listed in this manual could result in potential mild personal injury.



Indicates risks unrelated with personal injury (such as property damage).

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ZOOMLION

Operation and Safety Manual

Section 1 Safety Precautions



SECTION 1 SAFETY PRECAUTIONS

1.1 General

To Owners/Users/Operators:

Zoomlion appreciates your choice of our machine for your application. Our number one priority is user safety, which is best achieved by our joint efforts. The following requirements need to be adhere to for the purpose of safety operating:

- a) Obey all user rules, job site regulations and governmental regulations.
- b) Read, understand and obey all operating instructions on the machine and in this manual.
- c) Keep good safety operating convention.
- d) Allow only those authorized and qualified personnel to operate the machine under the supervision of an experienced and qualified operator.
- e) An operator must not operate the machine if he has any doubts.

Zoomlion appreciates your choice of our machine for your application.

1.2 Pre-operation



Failure to comply with the safety precautions listed in this manual could result in personal injury or death.

An operator must not operate the machine, only if:

- a) He has learned and practiced the principles of safe machine operation contained in this operational manual.
 - 1) Avoid hazardous situations.
 - 2) Be aware of safety rules before further operation.
 - 3) Perform a pre-operation inspection at all times.
 - 4) Implement functional test before operating the machine at all times.
 - 5) Inspect job site.
 - 6) Only use the machine as it was intended.
- b) Read, understand and obey the manufacturer's instructions and safety rules—safety and operator's manuals and machine decals.
- c) Read, understand and obey employer's safety rules and worksite regulations.

- d) Read, understand and obey all applicable governmental regulations.
- e) The operator is properly trained to safely operate the machine.

1.3 Hazard Classification

Decals on this machine use symbols, color coding, and signal words to identify the following:



Safety alert symbol—used to alert you to potential personal injury hazards. Obey all safety messages that follow this symbol to avoid possible injury or death.

▲ DANGER

Indicates a hazardous situation which, if not avoided, will result in death or serious injury. This decal will have a red background.

▲ WARNING

Indicates a hazardous situation which, if not avoided, could result in death or serious injury. This decal will have an orange background.

▲ CAUTION

Indicates a hazardous situation which, if not avoided, could result in minor or moderate injury. This decal will have a yellow background.

NOTICE

Indicates a property damage message. This decal will have a blue background.

1.4 Intended Use

This machine is intended to be used only to lift personnel, along with their tools, and materials to an aerial work site.

1.5 Safety Alert Symbols and Maintenance

Replace any missing or damaged safety signs. Keep operator safety in mind at all times. Use mild soap and water to clean safety signs. Do not use solvent-based cleaners because they may damage the safety sign material.

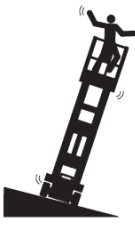
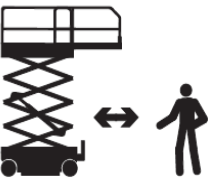
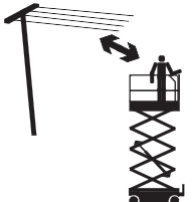
				
Read operational manual	Read service manual	Crushing hazard	Crushing hazard	Collision hazard
				
Tip-over hazard	Tip-over hazard	Tip-over hazard	Tip-over hazard	Electrocution hazard
				
Electrocution hazard	Explosion hazard	Fire hazard	Burn hazard	Dermohemia hazard
				
Engage safety arm	Keep away from moving parts	Keep clear of outriggers and tires	Move machine to level ground	Close chassis tray
				
Lower the platform	Do not set up where it cannot be leveled with outriggers	Maintain required clearance	Compartment maintenance implement by trained and authorized personnel only	Use a piece of cardboard or paper to search for leaks

Figure 1-1 Symbol and hazard pictorials definitions

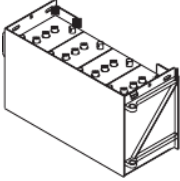
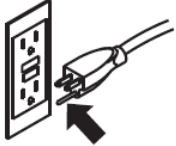
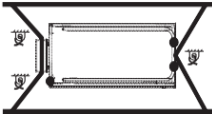
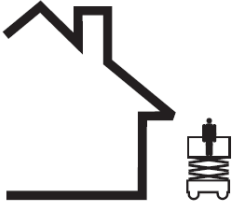
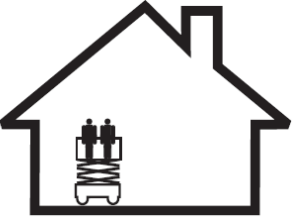
				
Batteries used as counterweights	Chock the wheels	Release brakes	3-wire grounded AC power supply only	Replace damaged wire and cable
				
Wheel load	Transporting diagram	Tie-down points	Lanyard anchorage points	No smoking
				
Voltage rating for power to platform	Pressure rating for air line to platform	Manual force	Wind speed	
				
Maximum capacity	Outdoor		Indoor	
				
Smashing hazard	Hold the guardrail when lowering		Outrigger load	

Figure 1-1 Symbol and hazard pictorials definitions

1.6 Safety Operation

1.6.1 Operator safety

Personal fall protection equipment (PFPE) is required when operating this machine. If PFPE is required in job site or in operator's manual, following rules should be complied with:

All PFPE must comply with applicable governmental regulations, and must be inspected and used in accordance with the PFPE manufacturer's instructions.

1.6.2 Workplace safety

1.6.2.1 Electric shock hazard

⚠ DANGER

This machine is not electrically insulated and will not provide protection from contact with or proximity to electrical current.

- a) Obey all local and governmental regulations regarding required clearance from electrical power lines. Keep required clearance listed in Table 1-1.



Table 1-1 Required clearance

No.	Voltage	Required clearance
1	0 to 50KV	3.05 m/10ft
2	50 to 200KV	4.60 m/15ft
3	200 to 350KV	6.10 m/20ft
4	350 to 500KV	7.62 m/25ft
5	500 to 750KV	10.67m/35ft
6	750 to 1000KV	13.72m/45ft

- b) Allow for platform movement, electrical line sway or sag, and beware of strong or gusty winds.
- c) Keep away from the machine if it contacts energized power lines. Personnel on the ground or in the platform must not touch or operate the machine until energized power lines are shut off.



- d) Do not operate the machine during lightning or storms.
- e) Do not use the machine as a ground for welding.

1.6.2.2 Tip-over hazard

⚠ DANGER

- a) Occupants, equipment and materials shall not exceed the maximum platform capacity.

Table 1-2 Rated load

Model	Maximum Capacity	Capacity on Platform Extended
ZS1218ERT	350kg /770 lbs	113kg /250 lbs

- b) Do not elevate the platform unless the machine is on firm level ground.



- c) Do not depend on the tilt alarm as a level indicator. The tilt alarm sounds on the chassis only when the machine is on a severe slope.

If the tilt alarm sounds: use extreme caution to lower the platform with the emergency lowering pull rod. Move the machine to a firm, level surface before lifting.

- d) Do not drive over 0.5mph (0.8km/h) with the platform raised.

Outdoor use: do not raise the platform when wind speeds may exceed 12.5 m/s (28 mph). Lower the platform and stop operating the machine if the wind speed exceeds 12.5 m/s (28 mph).

Indoor use: do not exceed the rated values of allowable manual force and maximum occupants listed in Table 1-3.

Table 1-3 Maximum allowable manual force

Model	Manual force	Maximum occupants
ZS1218ERT	400N/90 lbs force	3(Indoor)/2 (Outdoor)

- e) Do not operate the machine in strong or gusty winds. Do not increase the surface area of the platform or the load. Increasing the area exposed to the wind will decrease machine stability.



- f) Use extreme care and slow speeds while driving the machine in the stowed position across uneven terrain, debris, unstable or slippery surfaces and near holes and drop-offs.



- g) Do not drive the machine on or near uneven terrain, unstable surfaces or other hazardous conditions with the platform raised or extended.
- h) The following operations are prohibited at any circumstances:
- 1) Push the machine or other objects with the platform.
 - 2) Contact adjacent structures with the platform.
 - 3) Tie the platform to adjacent structures.
 - 4) Place loads outside the platform perimeter.
 - 5) Operate the machine with the chassis trays open.

- 6) Push off or pull toward any object outside of the platform.



- i) Do not modify or alter an aerial work platform without prior written permission from the manufacture.
 - 1) Do not alter or disable the limit switches.
 - 2) Do not alter or disable machine components that in any way affect safety and stability.
 - 3) Do not replace items critical to machine stability with items of different weight or specification.
 - 4) Mounting attachments for holding tools or other materials onto the platform, toeboards, or guard rail system can increase the weight in the platform and the surface area of the platform or the load.
- j) Do not use lead acid or lithium-ion batteries that weigh less than the original equipment. Batteries are used as counterweight and are critical to machine stability.
- k) Do not place or attach fixed or overhanging loads to any part of this machine.
- l) Do not place ladders or scaffolds in the platform or against any part of this machine.



- m) Do not transport tools and materials unless they are evenly distributed and can be safely handled by person(s) in the platform.
- n) Do not use the machine on a moving or mobile surface or vehicle. Be sure the tires are in good condition and the lug nuts tightened, besides the opening pin installed in the right position.

1.6.2.3 Crush hazard

⚠ DANGER

- a) Keep hands and limbs out of scissors.

- b) Do not work under the platform or in the scissor links without the safety arm in place.
- c) Use common sense and planning when operating the machine with the controller from the ground. Maintain safe distances between the operator, the machine and fixed objects.

1.6.2.4 Danger of operation on the slope

⚠ DANGER

Do not drive the machine on slopes that exceed its longitudinal and lateral ratings. Longitudinal and lateral ratings apply to machines in the stowed position.

Table 1-4 Longitudinal and lateral ratings for stowed position

Model	Maximum longitudinal rating for the Stowed Position	Maximum lateral Rating for the Stowed Position
ZS1218ERT	40% (22°)	40% (22°)

1.6.2.5 Fall off hazard

⚠ DANGER

The guard rails system provides fall protection. If occupant(s) of the platform are required to wear personal fall protection equipment (PFPE) due to job site or employer rules, PFPE and its use shall be in accordance with the PFPE manufacturer's instructions and applicable governmental requirements. Use approved lanyard attachment point provided.

- a) Do not sit, stand or climb on the platform guard rails. Maintain a firm footing on the platform floor at all times.
- b) Do not climb down from the platform when raised.



- c) Keep the platform floor clear of debris.
- d) Do not enter or exit the platform unless the machine is in the stowed position.
- e) Attach the platform entry chain or close the entry gate before operating.
- f) Do not operate the machine unless the guard rails are properly installed and the entry is secured for

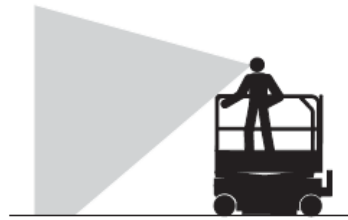
operation.

1.6.2.6 Collision hazard

⚠ DANGER

No stunt driving or horseplay while operating a machine.

- a) Be aware of limited sight distance and blind spots when driving and operating.



- b) Be aware of extended platform position when moving the machine.
- c) Be sure the machine is on a level surface or secured before releasing the brake.
- d) Operators must comply with employer, job site, and governmental rules regarding use of personal protective equipment.
- e) Check the work area for overhead obstructions or other possible hazards.



- f) Be aware of crushing hazards when grasping the platform guard rail.



- g) Observe and use the color-coded direction arrows on the platform controls and the platform decal plate for drive and steer functions.
- h) Do not lower the platform unless the area below is clear of personnel and obstructions.

起
重
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说
明

- i) Limit travel speed according to the condition of the ground surface, congestion, slope, location of personnel, and any other factors which may cause collision.



- j) Do not operate a machine in the path of any crane or moving overhead machinery unless the controls of the crane have been locked out and/or precautions have been taken to prevent any potential collision.

1.6.2.7 Burn hazard

⚠ DANGER

- a) Liquid or gas burn hazard.
 - 1) Do not operate a machine with a hydraulic oil or air leak. An air leak or hydraulic leak can penetrate and/or burn skin.
 - 2) Batteries contain acid. Always wear protective clothing and eye wear when working with batteries.



- 3) Avoid spilling or contacting battery acid. Neutralize battery acid spills with baking soda and water.
- 4) Do not expose the battery or charger in water or rain while charging.

⚠ DANGER

b) Electric shock or burn hazard.

- 1) Conduct daily check with wires and cables.



- 2) Change damaged items prior to operation. Avoid contact with battery terminals. Remove all rings, watches and jewelry.

- 3) Connect the battery charger to a grounded, AC 3-wire electrical outlet only.

1.6.2.8 Explosion and fire hazard

⚠ DANGER

- a) Do not operate the machine or charge the battery in hazardous locations or locations where potentially flammable or explosive gases or particles may be present.
- b) Keep sparks, flames, and lighted tobacco away from batteries. Batteries emit explosive gas.



- c) Swing out the charger tray when charging indoor for cooling.
- d) Do not use tools which could produce flames to contact battery terminals or cable clamp.
- e) Keep lithium-ion battery away from heat.
- f) Do not over charge or over discharge the lithium-ion battery.
- g) If there is heat, deformation, liquid leakage, odor or smoke when charging, stop using the lithium-ion battery and place the battery in an open place away from the crowd.
- h) It is strictly forbidden to immerse the lithium-ion battery in water, acid, alkaline and salt solution. Avoid rain.

1.6.2.9 Machine damage hazard

⚠ DANGER

- a) Do not use a damaged or malfunctioning machine. Conduct a thorough pre-operation inspection of

the machine and test all functions before each work shift.

- b) Immediately tag and remove from service a damaged or malfunctioning machine.
- c) Be sure all maintenance has been performed as specified in this manual and the appropriate Zoomlion service manual.
- d) Be sure all decals are in place and legible.
- e) Be sure operator's, safety, and responsibilities manuals are complete, legible, and in the storage container located on the machine.

1.6.2.10 Component damage hazard

⚠ DANGER

- a) Do not use a charger other than the ZOOMLION charger.
- b) When lifting the battery, please arrange the appropriate number of people and appropriate use and lifting methods, and lock the battery after each use.
- c) Do not use the machine as ground wire during welding;
- d) Do not use the machine in places where there may be strong magnetic fields.

ZOOMLION

Operation and Safety Manual

**Section 2 Machine Components
and Controls**



SECTION 2 MACHINE COMPONENTS AND CONTROLS

2.1 Machine Components

ZS1218ERT Components

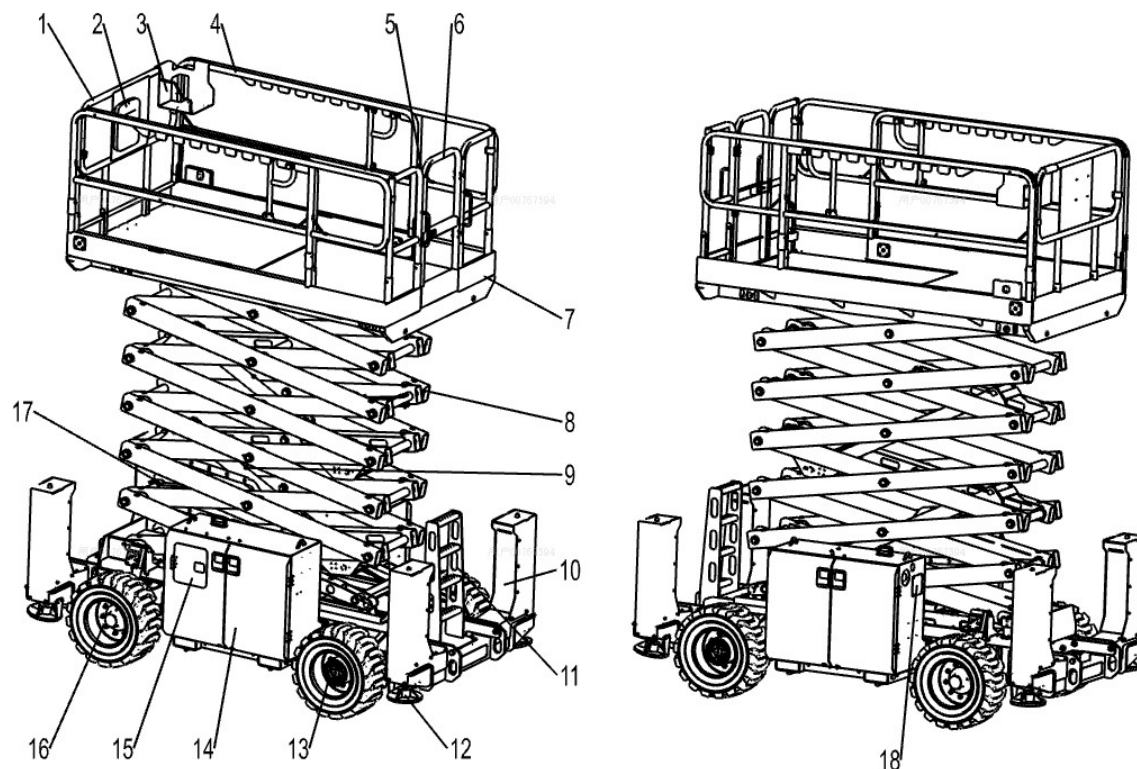


Figure 2-1 Components 1

Table 2-1 Component Instruction 1

No.	Item	No.	Item	No.	Item
1	Extension deck	7	Fixed platform	13	Non-steering wheel
2	Manual Storage Container	8	Scissor arm	14	Hydraulic tank
3	Platform controller	9	Lifting cylinder	15	Ground control
4	Platform guard rails	10	Outriggers	16	Steering wheel
5	Lanyard anchorage points	11	Entry ladder	17	Safety arm
6	Platform entry gate	12	Outrigger foot pad	18	Charger

2.2 Machine Controller

CAUTION

The manufacturer has no direct control over machine application and operation. The user and operator are responsible for conforming with good safety practices.

2.2.1 Ground controller

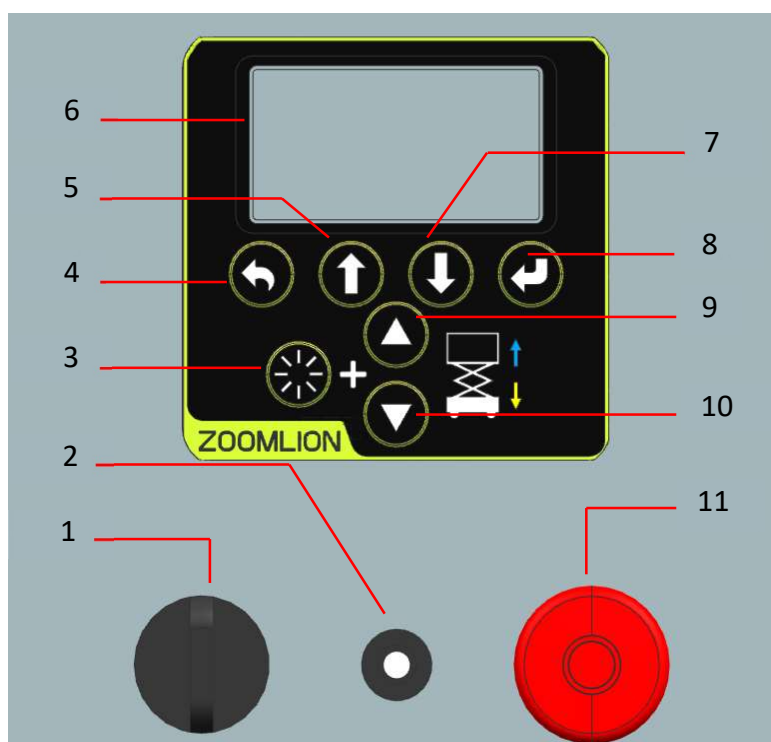


Figure 2-4 Ground control panel

Table 2-4 Ground control panel instruction

No.	Item	No.	Item
1	Key switch	7	Menu down button
2	10A breaker	8	Menu enter button
3	Lifting enable button	9	Platform up button
4	Menu escape button	10	Platform down button
5	Menu up button	11	Emergency stop switch
6	LCD diagnostic readout		

a) Key switch

Turn the key switch to the platform position and the platform controller will run. Turn the key switch to the off position and the machine will turn off. Turn the key switch to the ground position and the ground controller will run.

b) Breaker

When the line is overloaded or short-circuited, the circuit breaker can quickly cut off the circuit and play a protective role. After troubleshooting, press the circuit breaker button to reconnect the line.

c) Lifting enable button

Press this button to activate the platform lift function.

d) Menu exit button

After the ground controller enters the menu setting interface, press the menu exit button to exit the current menu interface and return to the upper-level menu.

e) Menu up button

After the ground controller enters the menu setting screen, you can switch between the cursor and the display screen up by pressing this button.

f) LCD display

The display screen of the ground controller can display the working condition, fault, menu setting interface, etc.

g) Menu down button

After the ground controller enters the menu setting screen, press this button to switch between the cursor and the display screen.

h) Menu enter button

After the ground controller enters the menu setting screen, press this button to enter the next level menu, or save the menu Settings.

i) Platform up button

Press the lift enable button + Platform up button at the same time, and the platform will rise.

j) Platform down button

Press the lift enable button + Platform down button at the same time to lower the platform.

k) Emergency stop switch

Push the emergency stop switch inward to the off position to stop all functions. Pull the emergency stop switch to the on (on) position to operate the machine.

2.2.2 Platform control unit

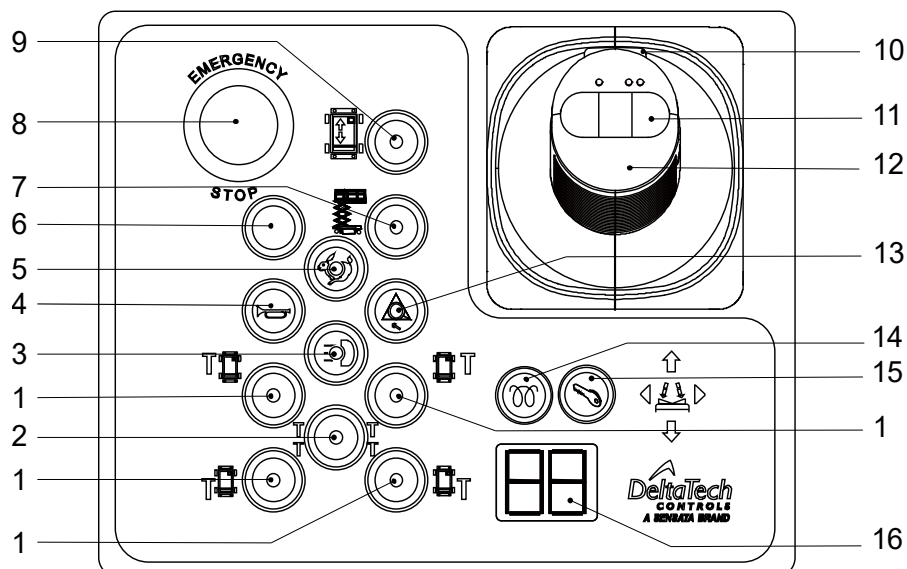


Figure 2-5 Platform control panel

Table 2-5 Platform control panel instruction

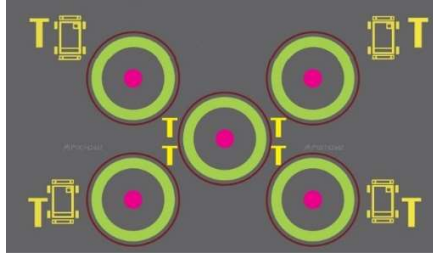
No.	Item	No.	Item
1	Outrigger function enable button with indicator	9	Drive function switch
2	Outrigger auto-leveling button	10	Function enable switch
3	Light button	11	Thumb rocker switch with steering function
4	Horn button	12	Proportional control joystick
5	Reserve	13	Tilting button with indicator
6	Reserve	14	Reserve
7	Lifting function enable button with indicator	15	Reserve
8	Red Emergency Stop button	16	Digital display

a) Outrigger start button and outrigger auto-leveling button with indicators

The button in the center of the figure below is the automatic outrigger leveling button. When this button is pressed, the indicator light of the lift function enable button and the drive function switch will light up at the same time, and the four outriggers can be activated at the same time to realize the automatic chassis leveling or retracting of the outrigger.

The left front, right front, left rear, and right rear outrigger function buttons enable the left front, right

front, left rear, and right rear outrigger functions respectively. When you press the enable button for one or more outrigger functions, the indicators of the enable button for the lift function and the switch for the drive function light up at the same time to activate the extension or retraction functions for one or more outrigger functions.



b) Light button

Optional. If the lighting is equipped, press this button and the lighting will work normally.

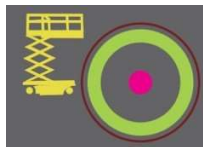
c) Horn button

Press the horn button and the horn will sound. Release horn button, horn does not sound.



d) Lifting enable button with indicator

Press this button to activate the lifting function. If no operation is performed after 7 seconds, press the button again to activate this function.

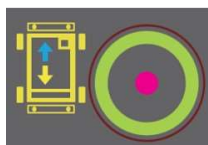


e) Red Emergency Stop button

Push the red "Emergency Stop" button inward to the off (Off) position to stop all functions. Pull the red "Emergency Stop" button to the on (on) position to operate the machine.

f) Drive function switch

Press this button to activate the driver function. If no operation is performed after 7 seconds, press the button again to activate this function.



g) Function enable button

Lifting function: First press the lift function button, then press and hold the function enable switch on the platform control lever to enable the lift function.

Driver function: Press the driver function button first, and then press and hold the function enable switch on the platform control lever to enable the driver function.

h) Thumb rocker switch for steering

Press the drive function button and then the left side of the thumb rocker, and the machine will turn in the direction indicated by the blue triangle on the platform control panel. Press the right side of the thumb rocker and the machine will turn in the direction indicated by the white triangle on the platform control panel.



i) Proportional control joystick

Lift function: Press the lift function button first, then press and hold the function enable switch on the platform control lever to enable the lift function. Move the control handle in the direction indicated by the blue arrow and the platform will rise. Move the control handle in the direction indicated by the white arrow and the platform will drop.

Driver function: Press the driver function button first, and then press and hold the function enable switch on the platform control lever to enable the driver function. Move the control handle in the direction indicated by the blue arrow on the control panel, and the machine will move in the direction indicated by the blue arrow. Move the control handle in the direction indicated by the white arrow on the control panel, and the machine will move in the direction indicated by the white arrow.

j) Machine tilting button with indicator

Press this button to activate the slow drive function. When driving slowly, the indicator light will light up. This feature should be selected in drive mode.



k) Digital display

Displays the current battery level or fault code.

ZOOMLION

Operation and Safety Manual

Section 3 Machine Inspection



SECTION 3 MACHINE INSPECTION

3.1 General

⚠ DANGER

An operator must not operate the machine, only if he has learned and practiced the principles of safe machine operation contained in this operational manual.

- a) Avoid hazardous situation.
- b) Perform a pre-operation inspection at all times.

Know and understand the pre-operation inspection before going on to the next section.

- c) Conduct functional test before usage.
- d) Inspect job site.
- e) Only use the machine as it was intended.

3.1.1 Pre-operation inspection fundamentals

- a) It is the responsibility of the operator to perform a pre-operation inspection and routine maintenance.
- b) The pre-operation inspection is a visual inspection performed by the operator prior to each work shift. The inspection is designed to discover if anything is apparently wrong with a machine before the operator performs the function tests.
- c) The pre-operation inspection also serves to determine if routine maintenance procedures are required. Only routine maintenance specified in this manual may be performed by the operator.
- d) Refer to the list on the next page and check each of the items.
- e) If damage or any unauthorized variation from factory delivered condition is discovered, the machine must be tagged and removed from service.
- f) Repairs to the machine may only be made by a qualified service technician, according to the manufacturer's specifications. After repairs are completed, the operator must perform a pre-operation inspection again before going on to the function tests.
- g) Scheduled maintenance inspections shall be performed by qualified service technicians according to the manufacturer's specifications and the requirements listed in the responsibilities manual.

3.1.2 Pre-operation inspection

- a) Be sure operation, safety, and responsibilities manuals are complete, legible, and in the storage container located on the machine.
- b) Be sure all decals are in place and legible. See Inspection section.
- c) Check for hydraulic oil leaks and proper oil level. Add oil if needed. See Maintenance section.

- d) Check the following components or areas for damage, improperly installed, or missing parts and unauthorized modification:
 - 1) Electrical components, wiring, and electrical cables.
 - 2) Hydraulic hoses, fittings, cylinders, manifolds.
 - 3) Drive motors.
 - 4) Wearing pads.
 - 5) Tires and wheels.
 - 6) Alarm and indicator (if equipped).
 - 7) Nuts, bolts and other fasteners.
 - 8) Brake release components.
 - 9) Safety arm.
 - 10) Platform extension.
 - 11) Scissor pin and retaining fasteners.
 - 12) Platform joystick.
 - 13) Outrigger hosing and foot pad (if equipped).
 - 14) Battery pack and connections.
 - 15) Ground strap.
 - 16) Platform entry chain or gate.
 - 17) Platform overload components.
 - 18) Pothole guards.
 - 19) Lanyard anchorage points.
 - 20) Check the machine for:
 - ① Cracks in welds or structural components
 - ② Dents or damage.
 - ③ Rust, corrosion, or oxidation.
- e) Be sure that all structural and other critical components are present and all associated fasteners and pins are in place and properly tightened.
- f) Be sure that the engine trays are closed and latched and the batteries are properly connected.
- g) Be sure that the hood is in place and locked after inspection.

⚠ CAUTION

If the platform must be raised to inspect the machine, make sure the safety arm is in place. See **Operating Instruction** section.

3.2 Function Test

3.2.1 Function test fundamentals

- a) The function tests are designed to discover any malfunctions before the machine is put into service. The operator must follow the step-by-step instructions to test all machine functions.
- b) A malfunctioning machine must never be used. If malfunctions are discovered, the machine must be tagged and removed from service. Repair to the machine may only be made by a qualified service technician, according to the manufacturer's specifications.
- c) After repairs are completed, the operator must perform a pre-operation inspection again before putting the machine into service.

⚠ DANGER

An operator must not operate the machine, only if he has learned and practiced the principles of safe machine operation contained in this operational manual.

- a) Avoid hazardous situations.
- b) Perform a pre-operation inspection at all times.
- c) Conduct functional test before usage.

Know and understand the pre-operation inspection before going on to the next section.

- d) Inspect job site.
- e) Only use the machine as it was intended.

3.2.2 Turn on the Power Switch

Pull out the red main power switch to on position which located on the side of battery container.



3.2.3 Ground Function Test

a) Test emergency stop.

- 1) Select a test area that is firm, level and free of obstruction.
- 2) Be sure the batteries are connected.
- 3) Pull out the platform and ground red Emergency Stop Button to the on position.
- 4) Turn the key switch to electronic control.
- 5) Observe the diagnostic LED readout on the platform controls. The LED should look like the picture below:



- 6) Observe the diagnostic LCD readout on the electronic controls. The LCD should display the model and hour meter:
- 7) Push in the ground red Emergency Stop Button to off position.
- 8) Result: No function should operate.
- 9) Pull out the red Emergency Stop Button to the on position.

b) Test the Up/Down functions.

This machine uses flashlights and buzzer warning.

Descent alarm: the flashlight illuminates 60 times per minute. The descent alarm sounds at 60 beeps per minute.

Descent delay alarm: the flashlight illuminates 120 times per minute. The descent alarm sounds at 120 beeps per minute.

When the machine is not level: the flashlight illuminates 120 times per minute. The descent alarm sounds at 120 beeps per minute.

- 1) Do not press the lift function enable button.
- 2) Press the platform up or down button.

Result: the lift function should not operate.

- 3) Do not press the platform up or down button.
- 4) Press the lift function enable button.

Result: the list function should not operate.

- 5) Press and hold the lift function enable button and the platform up button.

Result: the platform should raise.

- 6) Press and hold the lift function enable button and platform down button.

Result: the platform should lower.

The minimum distance between platform and ground should reach 2.55m. Descent delay light will illuminate and alarm will sound. Be sure there is no personnel or obstructions under the platform when lowering. Release and move the control handle to continue the lowering operation.

- c) Test emergency lowering.

- 1) Activate the up function by pressing the lift enable button and platform up button, and raise the platform approximately 60cm /2ft.
- 2) Pull the emergency lowering knob located behind the entry ladder.

Result: platform should lower. Descent alarm will not sound.

3.2.4 Platform function test

- a) Test emergency stop.

- 1) Select a test area that is firm, level and free of obstruction.
- 2) Turn the key switch to platform control.
- 3) Push in the platform red Emergency Stop Button to the off position.

Result: no function should operate.

- b) Test the horn.

- 1) Pull out the red Emergency Stop to the on position.
- 2) Press the horn button.

Result: the horn should sound.

- c) Test the Function Enable Switch and Up/Down function.

- 1) Do not hold the function enable switch on the control handle.
- 2) Slowly move the control handle in the direction indicated by the blue arrow, then in the direction indicated by the white arrow.

Result: no function should operate.

- 3) Press the lift function enable button.
- 4) Wait seven seconds for the lift function to time out.



- 5) Slowly move the control handle in the direction indicated by the blue arrow, then in the direction indicated by the white arrow.

Result: the lift function should not operate.

- 6) Press the lift function enable button, the indicator lights up.
- 7) Press and hold the function enable switch on the control handle when indicator lighting, and slowly move the control handle in the direction indicated by the blue arrow.

Result: the platform should raise.

- 8) Release the control handle.

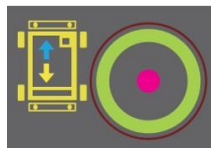
Result: the platform should stop raising.

- 9) Press and hold the function enable switch on the control handle when indicator lighting, and slowly move the control handle in the direction indicated by the white arrow.

Result: the platform should lower.

- d) Test drive function button.

- 1) Press the drive function button, then the indicator lights up.



- 2) Wait seven seconds for the drive function to time out.
- 3) Slowly move the control handle in the direction indicated by the blue arrow, then in the direction indicated by the white arrow.

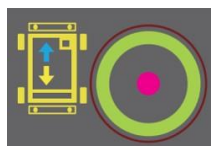
Result: no function should operate.

- e) Test the steering.

CAUTION

When performing the steer and drive function tests, stand in the platform facing the steer end of the machine.

- 1) Press the drive function button, then the indicator lights up.



- 2) Press and hold the functional enable switch on the control handle when lighting.
- 3) Press the thumb rocker switch on top of control handle in the direction indicated by the blue

triangle on the control panel.

Result: the steer wheels should turn in the direction indicated by the blue triangle.

- 4) Press the thumb rocker switch on top of the control handle in the direction indicated by the white triangle on control panel.

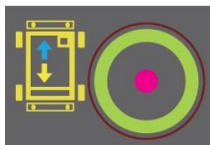
Result: the steer wheels move towards the direction indicated by the yellow triangle.

- f) Test driving and braking.

CAUTION

The brake must be able to hold the machine on any slope it is able to climb. This test is performed on the ground with a platform controller. Do not stand in the platform.

- 1) Press the drive function button, then the indicator lights up.



- 2) Press the function enable switch on the control handle when lighting.
- 3) Slowly move the control handle in the direction indicated by the blue arrow on the control panel until the machine begins to move, then return the control handle to the center position.

Result: the machine should move in the direction that the blue arrow points on the control panel, then come to an abrupt stop when the control handle is returned to the center position.

- 4) Slowly move the control handle in the direction indicated by the white arrow on the control panel until the machine begins to move, then return the control handle to the center position.

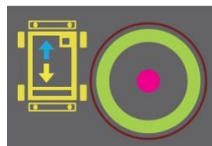
Result: the machine should move in the direction that the white arrow points on the control panel, then come to an abrupt stop when the control handle is returned to the center position.

- g) Test the tilt sensor operation.

- 1) Fully lower the platform.
- 2) Place a 5*10cm/2*4in or similar piece of wood under both wheels on the side and drive the machine up onto them.
- 3) Raise the platform approximately 3.2m from the ground.

Result: the platform should stop raising. The tilt alarm will sound and the light will illuminate. Platform controller LED readout displays LL, ground controller LCD displays LL: Machine Tilted.

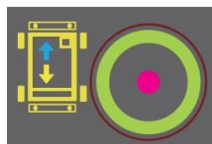
- 4) Press the drive function button, then the indicator lights up.



- 5) Press and hold the functional enable switch on the control handle.
- 6) Move the control handle in the direction indicated by the blue arrow, then move the control handle in the direction indicated by the white arrow.

Result: the drive function should not work in either direction.

- 7) Lower the platform and remove both pieces of wood.
- h) Test elevated drive speed.
 - 1) Raise the platform approximately 3.2m from the ground.
 - 2) Press the drive function button, then the indicator lights up.



- 3) Press and hold the functional enable switch on the control handle. Slowly move the control handle to full drive position.

Result: the maximum achievable drive speed with the platform raised should not exceed 0.5km/h.

- 4) If the drive speed with the platform raised exceeds 0.5km/h, immediately tag and remove the machine from service.
- i) Test oscillating system

Note: Perform this test on the ground with the platform controller, do not stand inside the platform.

- a) Test oscillating system (stowed position):

- 1) Maintain platform at the stowed position
- 2) Drive left wheel to turn to a slope at a height of 0.1m

Result: All four tires should be in close contact with the ground.

- 3) Drive right wheel to turn to a slope at a height of 0.1m

Result: All four tires should be in close contact with the ground.

Note: Check the display of the ground controller and require no fault code to be displayed.

- b) Test oscillating system (raised position, below the safe altitude):

- 1) Press the start button of the lifting function, and lift the platform to a height of 2-2.5m above

the ground with the handle

- 2) Drive the left steering wheel into the 0.1m deep pothole

Result: All four tires should be in close contact with the ground.

- 3) Drive the right steering wheel into the 0.1m deep pothole

Result: All four tires should be in close contact with the ground.

Note: Check the display of the ground controller and require no fault code to be displayed.

- c) Test oscillating system (raised position, above the safe altitude):

- 1) Press the start button of the lifting function, and lift the platform to a height of 3.2m above the ground with the handle

- 2) Drive the left steering wheel into the 0.1m deep pothole

Result: One of the tires doesn't touch the ground.

- 3) Drive the right steering wheel into the 0.1m deep pothole

Result: One of the tires doesn't touch the ground.

Note: Check the display of the ground controller and require no fault code to be displayed.

3.2.5 Turn off the main power switch

When the machine has not been used for a long time, press the red Main Power Switch on the side of the battery box to the off position to turn off the main power.



3.3 Workplace Inspection

⚠ DANGER

An operator must not operate the machine, only if: he has learned and practiced the principles of safe machine operation contained in this operational manual.

- a) Avoid hazardous situations.
- b) Perform a pre-operation inspection at all times.
- c) Implement functional test before operation the machine at all times.
- d) Inspect job site

Know and understand the workplace inspection before going on to the next section.

- e) Only use the machine as it was intended.

3.3.1 Workplace inspection fundamentals

Workplace inspection helps the operator determine if the workplace is suitable for safe machine operation.

It should be performed by the operator prior to moving the machine to the workplace.

It is operator's responsibility to read and remember the workplace hazards, then watch for and avoid them while moving, setting up, and operating the machine.

3.3.2 Workplace inspection

Be aware of and avoid the following hazards:

- a) Drop-offs or holes.
- b) Bumps, floor obstructions or debris.
- c) Sloped surfaces.
- d) Unstable or smooth surfaces.
- e) Overhead obstructions and electric lines.
- f) Hazardous locations
- g) Inadequate surface support to withstand all load forces imposed by the machine.
- h) Wind and weather conditions.
- i) Unauthorized personnel.
- j) Other possible unsafe conditions.

3.4 Decals Inspection

Use the lists and pictures below to verify that all decals are legible and in place.

Table 3-1 Decal 1 (ZS1218ERT)

NO.	Code	Item	Qty
1	00773807000402040	Label - Load of 350kg (outdoor)	1
2	007753070J0201080	Label – Max manual force (indoor)	1
3	00775207050402090	Label – Uninsulated (English)	1
4	007753070J0201020	Label – zoomlion 75	3
5	00773807000402070	Label – Colliding hazard	2
6	00775207050403040	Label – Safety rule and overturning (outdoor)	1
7	00773807000401030	Label - ZS1218ERT 75	2
8	00775207050403080	Label – Rated voltage of platform power	1
9	00773807000402060	Label – Main power switch operation instruction	1
10	00775207050403060	Label – Charger power	1
11	00773807000401070	Label – Battery box	1
12	00775207050401040	Label – Fork hole	4
13	00773807000401020	Label - ZS1218ERT 50	2
14	00775207050401010	Label – Anti-pinch	4
15	00773807000401010	label – Zoomlion 50	1
16	00773807000402050	Label – Outrigger load and keep away from	4
17	00773807000401040	Label – Wheel load of 1350kg	4
18	00775207050402010	Label – Smashing and safety arm	4
19	00775207050401050	Label – Lifting and tie up	4
20	00775207050403030	CE label	1
21	00775207050401030	IPAF label	1
22	00775207050402050	Label – Handrail position	3
23	00775207050401060	Label – Safety lanyard point	6
24	00775807010403050	Label – Yellow reverse	2
25	00773807000401090	Label – No high pressure water washing	2
26	00773807000402020	Label – Hydraulic tank	1
27	00775807000201060	Label – Key switch	1
28	00775207050401090	Label – Read instruction carefully	2
29	00773807000401080	Label – No stepping	2
30	00775207050402040	Label – Platform AC power (chassis)	1
31	00773807000402080	Label – No stepping (small)	2
32	00775807010403060	Label – Blue forward	2
33	00775207050403020	Label – Overturning hazard (chassis)	1
34	00773807000402030	Label – Smashing and outrigger tilting	1
35	007753070J0201090	Label – Emergency lowering	1

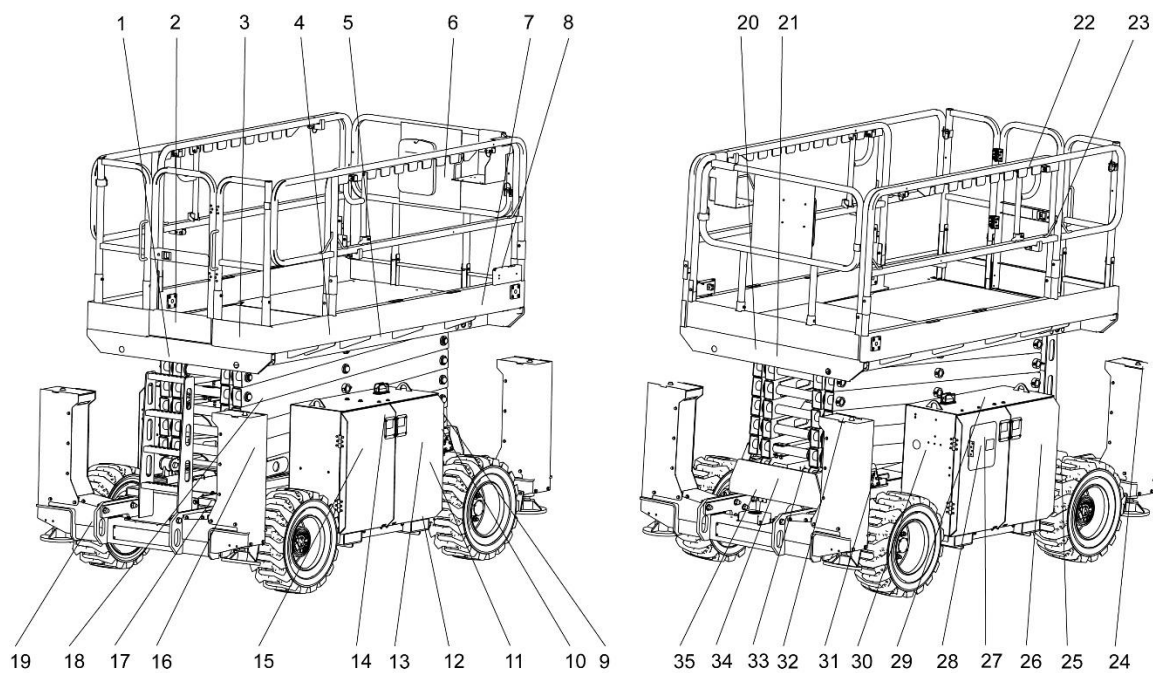


Figure 3-1 Decal position 1 (ZS1218ERT)

ZS1218ERT

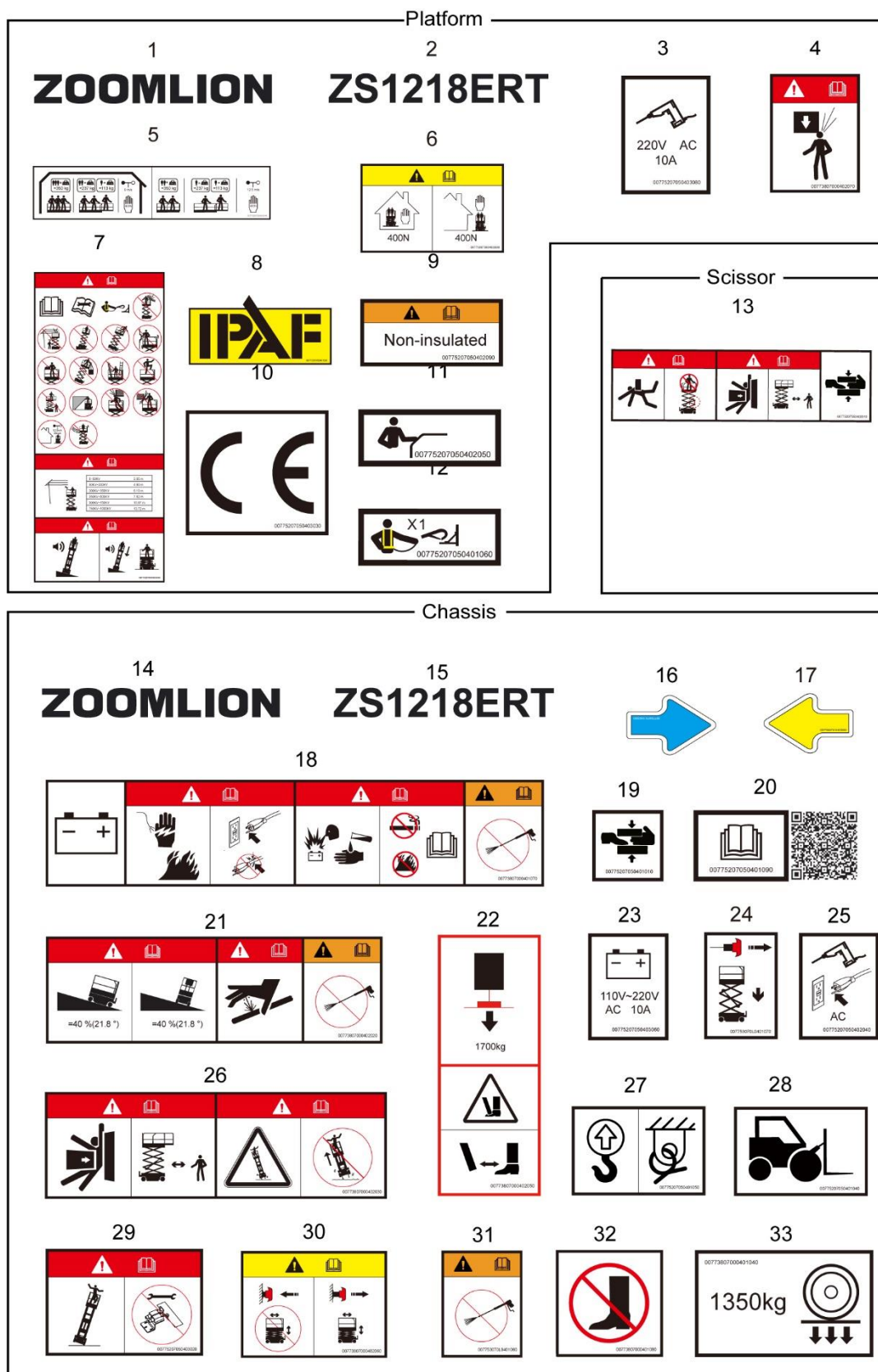


Figure 3-2 Decal

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Operation and Safety Manual

Section 4 Operation Instruction

SECTION 4 OPERATION INSTRUCTION

4.1 General

⚠ DANGER

An operator must not operate the machine, only if he has learned and practiced the principles of safe machine operation contained in this operational manual.

- a) Avoid hazardous situations.
- b) Perform a pre-operation inspection at all times.
- c) Implement functional test before operating the machine at all times.
- d) Inspect job site.
- e) Only use the machine as it was intended.

Fundamentals:

The Operating Instructions section provides instructions for each aspect of machine operation. It is the operator's responsibility to follow all the safety rules and instructions in the operator's safety and responsibilities manuals.

Using the machine for anything other than lifting personnel, along with their tools and materials, to an aerial work site is unsafe and dangerous.

Only trained and authorized personnel should be permitted to operate a machine. If more than one operator is expected to use a machine at different times in the same work shift, they must all be qualified operators and are all expected to follow all safety rules and instructions in the operator's, safety and responsibilities manuals. That means every new operator should perform a pre-operation inspection, function tests, and a workplace inspection before using the machine.

4.2 Machine Operation

4.2.1 Turn on/off main power switch

- a) Push in the red Power Switch to the off position at the battery box side to shut off machine power.
- b) Pull out the red Power Switch to on position to operate the machine.
- c) If the machine is out of service for a long time or in maintenance, turn off the Power Switch.



4.2.2 Emergency stop

- a) Push in the red Emergency Stop button to the off position at the electronic controls or the platform controls to stop all functions.
- b) Repair any function that operates when Power Switch and red Emergency Stop button are pushed in.

4.2.3 Emergency lowering

Pull the emergency lowering knob to lower the platform.

4.2.4 Operation after usage

- a) Select a safe parking location-firm level surface, clear of obstruction and traffic.
- b) Lower the platform.
- c) Turn the key switch to the off position and remove the key to secure from unauthorized use.
- d) Charge the batteries.

4.3 Operation from Ground

⚠ CAUTION

Maintain safe distances between the operator, the machine and fixed objects.

Notice the drive direction when using the controller.

4.3.1 Start electronic operation function

- a) Be sure the battery pack is connected before operating the machine.
- b) Turn the key switch to electronic control.
- c) Pull out both ground and platform red Emergency Stop buttons to the on position.

4.3.2 Adjust platform position

Press and hold both lifting enable button and platform up/down button on control panel to adjust platform position.

Drive and steer functions are not available from the electronic controls.

4.4 Operation from Platform

4.4.1 Start platform operation function

- a) Be sure the battery pack is connected before operating the machine.
- b) Turn the key switch to platform control.
- c) Pull out both ground and platform red Emergency Stop buttons to the on position.

4.4.2 Adjust platform position

- a) Press the lift function enable button.

CAUTION



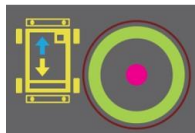
A round indicator below the lift function symbol will turn on.

If the control handle is not moved within seven seconds of pushing the lift button, the indicator below the lift function symbol will turn off and lift function will not operate.

- b) Press and hold function enable switch on handle while the lifting function indicator is on.
c) Move the control handle according to markings on the control panel.

4.4.3 Steering

- a) Press the drive function button.



A round indicator below the drive function symbol will turn on.

If the control handle is not moved within seven seconds of pushing the drive function button, the circle below the drive function symbol will turn off and drive function will not operate.

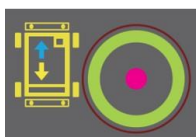
- b) Press and hold function enable switch on handle while the drive function indicator is on.
c) Press and hold steering thumb button on the top of handle for steering according to the symbols on control panel.



4.4.4 Drive

- a) Press the drive function button. A round indicator below the drive function symbol will turn on.

If the control handle is not moved within seven seconds of pushing the drive function button, the indicator below the drive function symbol will turn off and drive function will not operate.



- b) Press and hold function enable switch on handle while the drive function indicator is on.
c) Increase speed: slowly move the control handle off center.

- d) Decrease speed: slowly move the control handle toward center.
- e) Stop: return the control handle to center or release the function enable switch.
- f) Use the color-coded direction arrows on the platform controls and on the platform to identify the direction the machine will travel.
- g) Machine travel speed is restricted when the platform is raised.
- h) Battery condition will affect machine performance. Machine drive speed and function speed will drop when the battery level indicator is flashing.

4.4.5 Drive speed selection

Select the desired drive speed by control handle, usually, the machine is in the normal drive speed. Press the slow drive speed mode button, the circle below the button will turn on, slow drive speed mode is activated, the drive speed will be restricted even if the control handle moves to the max speed position.



4.4.6 Outrigger operation

Auto leveling:

- a) Place the machine below the intended work area;
- b) Press and hold the automatic leveling button, and the indicator light of the lifting function enable button and the driving function switch will be lit at the same time;
- c) Start the proportional control handle in the lowering direction, and the outrigger will extend;
- d) When the outrigger touches the ground, the indicator corresponding to the outrigger function start indicator will light up;
- e) Continue to extend the outrigger, the machine gradually levels, the machine level will make a prompt tone.
- f) When an outrigger touches the ground, the chassis drive function is prohibited.

Control a single outrigger:

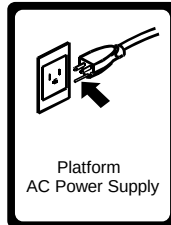
- a) Hold down one or more outrigger function start buttons, lift function enable button and drive function switch indicator light will be lit at the same time;

Start the proportional control handle according to the required direction, and realize the expansion and contraction of the outrigger to level the machine.

4.4.7 Platform AC power (AC 220V)

When using the platform AC power supply (if equipped with the machine), pull out the three-hole power

plug from the side of the machine box and connect it to the grounded AC three-wire power supply (AC220V) socket. The AC power socket on the platform can provide 220V AC power.



4.5 Operation on Slope

4.5.1 Driving on a slope

⚠ DANGER

- a) Determine the longitudinal and lateral ratings and slope of the slope under machine retract condition.

Table 4-1 Longitudinal rating

Maximum longitudinal rating in the stowed state			
	ZS1218ERT	40%	22°

Table 4-2 Lateral rating

Maximum lateral rating in the stowed state			
	ZS1218ERT	40%	22°

- b) Slope rating is limited by ground conditions and traction. Press the drive speed button to the fast drive speed mode.
- c) Measure the slope with a digital inclinometer OR use the following procedure. You will need: carpenter's rule, straight block (minimum length 1m/ 3ft 3in), tape measure.
- 1) Lay the piece of wood on the slope.
 - 2) At the downhill end, lay the level on the top of edge of the piece of wood and lift the end until the piece of wood is level.
 - 3) While holding the piece of wood level, measure the vertical distance from the bottom of the piece of wood to the ground. Divide the tape measure distance (rise) by the length of the piece of wood (run) and multiply by 100.

Example:

Block = 3.6m /12ft (144in),



Stroke=3.6m /12ft,

Rise = 0.3m /12in,

Grade $0.3\text{m} / 12\text{in} \div 3.6\text{m} / 144\text{in} = 0.083 \times 100 = 8.3\%$.

If the slope exceeds the maximum slope or side slope rating, then the machine must be winched or transported up or down the slope. See Transport and Lifting section.

4.5.2 Operating on a slope

⚠ DANGER

Determine the rating of the chassis working Angle when the machine is working longitudinally and laterally on the slope.

Table 4-3 Maximum chassis working Angle

Model	Longitudinal rating	Lateral rating
ZS1218ERT	3°	2°

If the slope exceeds the chassis working Angle rating, tilt alarm will sound, then the platform must be lowered carefully. Move the machine to a firm, level surface before lifting.

4.6 Traction

a) Notice (Electric-drive models):

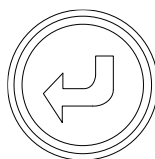
It is not recommended to tow the machine, except in the event of an emergency such as a machine malfunction or a total machine power failure.

b) Warning:

- 1) If the machine has any inclination, the wheel must be fully blocked before manually releasing the brake, otherwise it may cause injury or even death.
- 2) Runaway Hazard. The equipment does not have traction brakes and the towing vehicle must be able to control the equipment at all times. Do not conduct traction on the highway. Failure to do so could result in serious injury.
- 3) Maximum traction speed should not exceed 3.2km/h (2 mile/h), and the traction spacing should not exceed 18m /60 ft. Maximum traction slope should not exceed 25%.

c) Electronically released brake (Electric-drive models):

- 1) Pull out device main power switch, platform emergency stop button and ground emergency stop button.
- 2) Press and hold ECU Menu Entry Button.



- 3) Turn the key switch to ground control.
- 4) Until the Main Menu appears on the LCD reading device:
- 5) Scroll to Other by pressing the Menu Down Button:
- 6) Press Menu Entry Button;
- 7) Scroll to Brake Release by pressing the Menu Down Button:
- 8) Press the Menu Entry Button, Brake Is Released! on display, and the buzzer will alarm, indicating that the brake has been released.
- 9) After the equipment is towed, the key switch is turned off and the brake is restored.

4.7 Operation Code

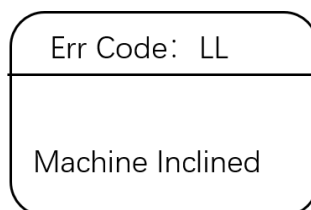
4.7.1 Operation indicator code

If the platform controls LED or electronic controls LCD diagnostic readout displays an operational indicator code such as LL, the fault condition must be repaired or removed before resuming machine operation. Push in and pull out the red Emergency Stop button to reset the system.

- a) LED Readout.



- b) LCD Readout.



- c) Operation Indicator Code:

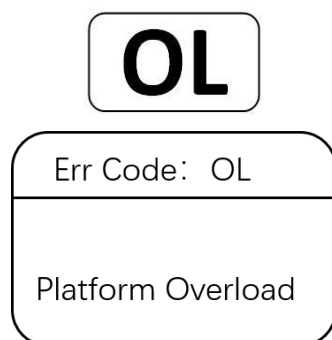
- 1) LL Off-Level.
- 2) OL Overload.
- 3) CH Chassis Mode Operation.

Refer to Zoomlion Maintenance manual for further information.

A code and a description of a code can also be viewed at the electronic controls LCD display.

4.7.2 Platform overload

If the platform controller LED diagnostic readout displays OL, and the electronic controller LCD diagnostic readout displays OL as well, it indicates platform overloaded, except for the descent function, other functions will stop. Alarm will sound.



- Press the red Emergency Stop button to off position.
- Remove load from platform.
- Pull out the Red Emergency Button to the on position to operate the machine.
- The reading device shows normal.

4.7.3 Battery level indicator

Use the LED diagnostic readout to determine the battery level.

When Low Charge appears on the platform controls LED display, the machine must be taken out of service and charged, otherwise all machine functions will be disabled.

Battery Level:90-100%	Battery Level:70%
Battery Level:50%	Battery Level:30%
Battery Level:20%	Battery Level:10%

4.8 Safety Arm and Guard Operation

4.8.1 How to use the safety arm

- a) Attempt to raise the platform to approximately 2.8m .
- b) Rotate the safety arm away from the machine and let it hang down.
- c) Lower the platform until the safety arm rests securely on the link. Keep clear of the safety arm when lowering the platform.

4.8.2 How to fold guard

The platform railing system consists of three fold down rail sections for the extension deck and three sections for the main deck. All six sections are held in place by four wire lock pins.

- a) Fully lower the platform and retract the platform extension.
- b) Ensure that the pull handle is in stowed condition;
- c) Remove the platform controls.
- d) From inside the platform, remove the two extension deck lock pins.
- e) Fold the front rail components. Keep hands free of pinch points.
- f) Replace the two moved pins back into each side rail bracket.
- g) Fold the rail components of each side. Keep hands free of pinch points.
- h) At the rear of the main deck, remove the two main deck lock pins.
- i) Carefully open the gate and exit the platform.
- j) Fold down the rear gate and entry side rails as one unit. Keep hands free of pinch points.
- k) Fold down the left and right side rails. Keep hands free of pinch points.
- l) Replace the two moved pins back into each side rail bracket.

4.8.3 How to raise guard

Follow the fold down instructions but in reverse order, ensuring all lock pins are in place and installed properly.

4.8.4 Operation after usage

- a) Select a safe parking location-firm level surface, clear of obstruction and traffic.
- b) Lower the platform.
- c) Turn the key switch to the off position and remove the key to secure from unauthorized use.
- d) Charge the batteries.

4.9 Battery and Charger Operation

⚠ CAUTION

Battery and Charger Instruction.

4.9.1 Observe and obey

- a) Do not use an external charger or booster battery.
- b) Charge the battery in a well-ventilated area.
- c) Use proper AC input voltage for charging as indicated on the charger.
- d) Use only a ZOOMLION authorized battery and charger.

4.9.2 Charging the battery

- a) Be sure the batteries are connected before charging the batteries.
- b) During the battery charging process, the battery tray should be kept closed and protected from rain.

4.9.3 Maintenance free battery

- a) Connect the battery charger to a grounded AC circuit.
- b) The charger will indicate when the battery is fully charged.

4.9.4 Standard battery

- a) Remove the battery vent caps and check the battery acid level. If necessary, add only enough distilled water to cover the plates. Do not add excessive distilled water before the charging process.
- b) Replace the battery vent caps.
- c) Connect the battery charger to a grounded AC circuit.
- d) The charger will indicate when the battery is fully charged.
- e) Check the battery acid level when the charging cycle is complete. Replenish with distilled water to the bottom of the fill tube. Do not overfill.

4.10 Lithium-ion Battery and Charger Operation

⚠ CAUTION

Lithium-ion Battery and Charger Instruction

4.10.1 Observe and obey

- a) When charging the lithium battery, please choose a power socket with sufficient load to charge the device, and the conventional ordinary socket (10A) can only charge one device;
- b) Do not over charge or over discharge the lithium-ion battery.

- c) If there is heat, deformation, liquid leakage, odor or smoke when charging, stop using the lithium-ion battery and place the battery in an open place away from the crowd.
- d) The battery is only suitable for the matching equipment, and the battery should not be used in other occasions.
- e) It is forbidden to use wire to directly short circuit the battery output;
- f) Do not use or store this product in environments such as corrosive, explosive, high temperature (heating, near fire or sun exposure).
- g) When charging, please use the special charger that comes with the device, and avoid charging in direct sunlight. Do not let children touch the charger in use.
- h) If the battery system is not used for a long time, it should be placed in a cool and dry environment (temperature below 30°C/86°F, humidity less than 90%) and charged at least once every three months.
- i) When the battery voltage is too low, it should be charged in time, otherwise the battery will be over-discharged and the machine will not move.
- j) Do not apply external force to the battery or drop it from high altitude.
- k) Keep the battery away from heat.
- l) Do not use an external charger or booster battery.
- m) Charge the battery in a well-ventilated area.
- n) Use proper AC input voltage for charging as indicated on the charger.
- o) Use only a ZOOMLION authorized charger.
- p) It is strictly forbidden for non-professionals to disassemble and modify the battery system. For repairs, please contact our after-sales personnel.

4.10.2 Charge lithium-ion battery

- a) Connect the battery charger to a grounded AC circuit.
- b) During the battery charging process, the battery tray should be kept closed and protected from rain.
- c) The charger will give instructions when the battery is fully charged.

4.11 Transport and Lift

4.11.1 Observe and obey

- a) ZOOMLION provides this securement information as a recommendation. The driver is solely responsible for ensuring that the machine is properly secured and the correct trailer is selected pursuant to CHINA Department of Transportation regulations, other localized regulations, and their company policy.

- b) ZOOMLION customers needing to containerize any lift or ZOOMLION product should source a qualified freight forwarder with expertise in preparing, loading and securing construction and lifting equipment for international shipment.
- c) Only qualified aerial lift operators should move the machine on or off the truck.
- d) The transport vehicle must be parked on a level surface.
- e) The transport vehicle must be secured to prevent rolling while the machine is being loaded.
- f) Be sure the vehicle capacity, loading surfaces and chains or straps are sufficient to withstand the machine weight. ZOOMLION lifts are very heavy relative to their size. See the serial label for the machine weight.
- g) Be sure the machine is on a level surface or secured before releasing the brake.
- h) Do not allow the rails to fall when the snap pins are removed. Maintain a firm grasp on the rails when the rails are lowered.
- i) Do not drive the machine on a slope that exceeds the uphill, downhill or side slope rating. See Driving on a Slope in the Operating Instructions section.
- j) If the slope of the transport vehicle bed exceeds the maximum slope rating, the machine must be loaded and unloaded using a winch as described in the brake release operation.

4.11.2 Transport

- a) Brake Release Operation:

- 1) Chock the wheels to prevent the machine from rolling.



- 2) Be sure the winch line is properly secured to the drive chassis tie points and the path is clear of all obstructions.
- b) After loaded:
 - 1) Chock the wheels to prevent the machine from rolling.
 - 2) Push the red Emergency Stop button at both ground and platform controls to the off position.
- c) Towing the machine is not recommended. If the machine must be towed, do not exceed 6km/h.
- d) Securing to Truck or Trailer for Transit:
 - 1) Always use the extension deck lock when the machine is transported. Turn the key switch to the off position and remove the key before transporting.
 - 2) Inspect the entire machine for loose or unsecured items. Use chains or straps of ample load capacity.

- 3) Use a minimum of 2 chains or straps. Adjust the rigging to prevent damage to the chains.

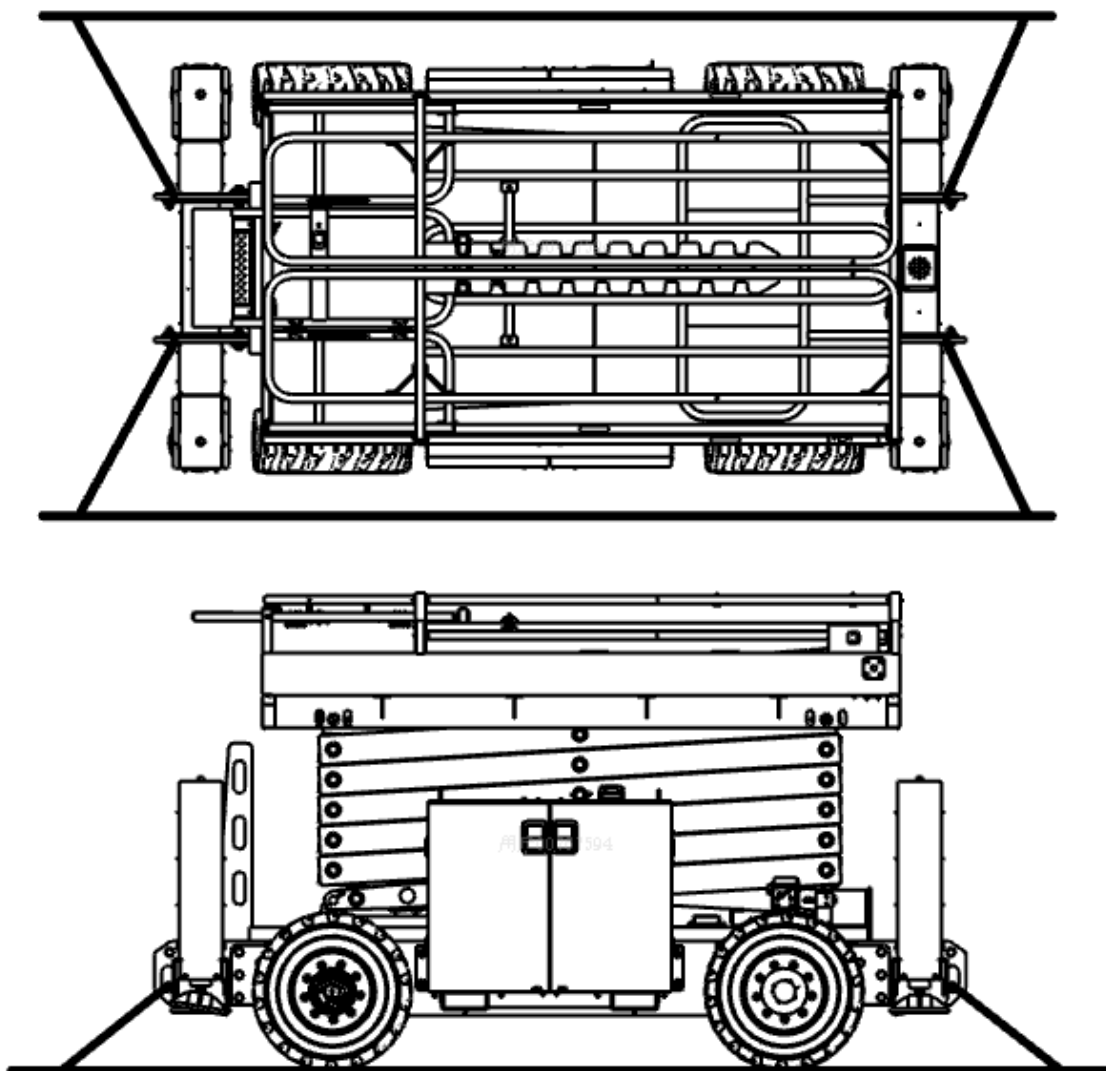


Figure 4-2 Tie down points

4.11.3 Lift

⚠ WARNING

- a) Observe and Obey:
 - 1) Only qualified riggers should rig and lift the machine.
 - 2) Only qualified forklift operators should lift the machine with a forklift.
 - 3) Be sure the crane capacity, loading surfaces and straps or lines are sufficient to withstand the machine weight. See the serial label for the machine weight.
- b) Lifting the machine with a Forklift:

- 1) Be sure the extension deck, controls and component trays are secured. Remove all loosen parts.
- 2) The platform must remain lowered during all loading and transport procedures.
- 3) Use the forklift pockets located on both sides of the ladder. See Figure 4-3.

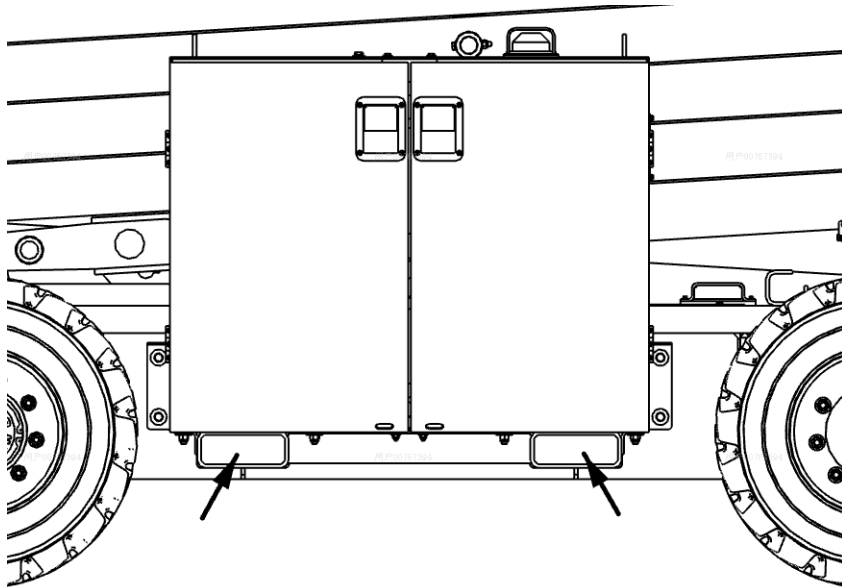


Figure 4-3 Forklift pockets

- 4) Align the forklift fork with the position of the forklift pockets and drive forward until the fork is fully inserted.
 - 5) Raise the machine 0.4m /1ft 4in and then tilt the forks back slightly to keep the machine secure.
 - 6) Be sure the machine is level when lowering the forks.
- c) Lifting Instruction:
- 1) Fully lowering the platform. Be sure the extension deck, controls and component trays are secured. Remove all loose items on the machine.
 - 2) Use Table 4-4 and Figure 4-4 to determine the center of gravity of the machine.
 - 3) Attach the rigging only to the designated lifting points on the machine. There are two 6cm holes on the front of the machine, and two holes on the rear of the machine for lifting.
 - 4) Adjust the rigging to prevent damage to the machine and to keep the machine level.

Table 4-4 Center of gravity

Model	X Axis	Y Axis
ZS1218ERT	103 cm	76 cm

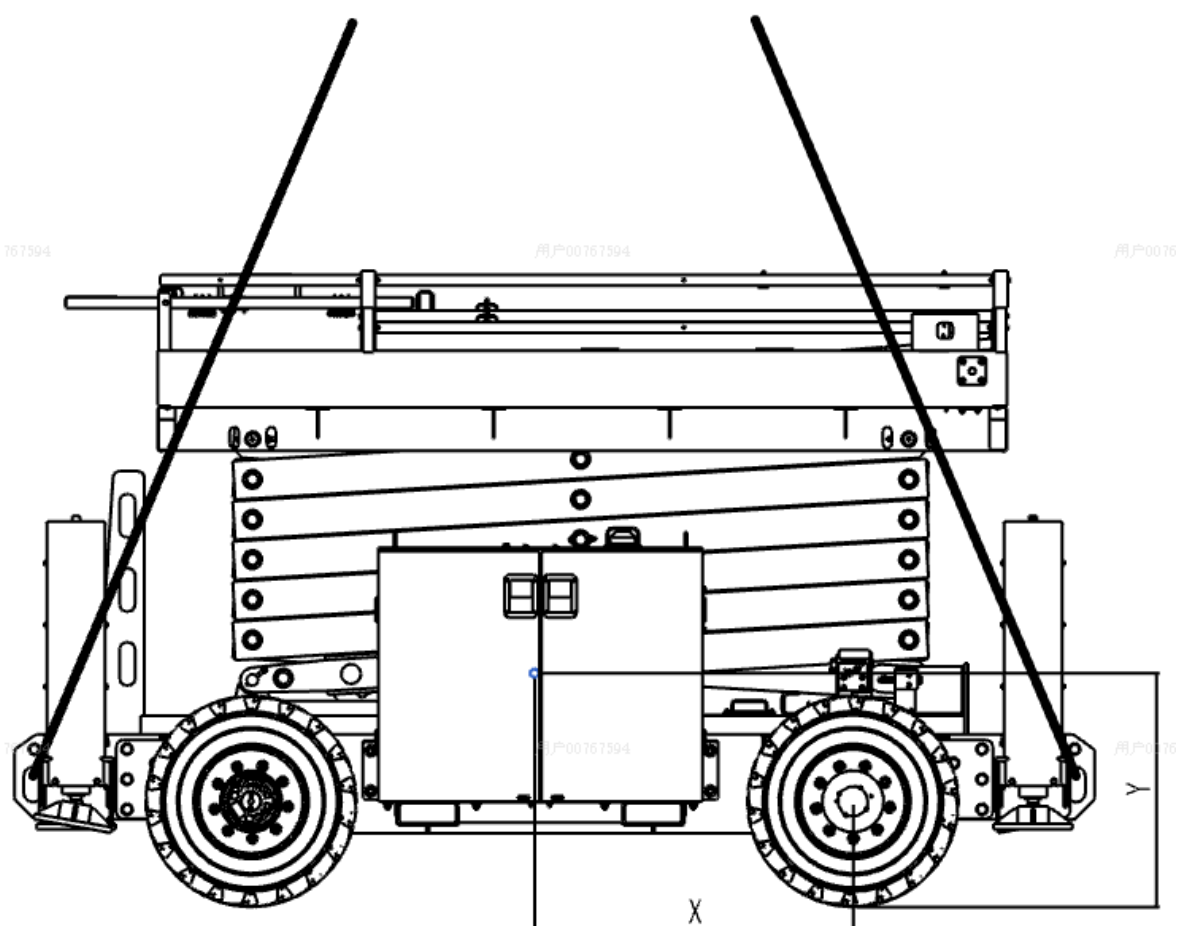


Figure 4-4 Center of gravity

ZOOMLION

Operation and Safety Manual

Section 5 Maintenance



SECTION 5 MAINTENANCE

5.1 General



Observe and Obey:

- a) Only routine maintenance items specified in this manual shall be performed by the operator.
- b) Scheduled maintenance inspections shall be performed by qualified service technicians, according to the manufacturer's specifications and the requirements listed in the responsibility's manual.
- c) Disposal of materials should be according to the regulations of government and relevant environmental protection administration.
- d) Use only ZOOMLION approved replacement parts. ZOOMLION assumes no responsibility for hazards occurred to equipment and personnel caused by the use of unauthorized parts.

5.1.1 Maintenance symbols legend

The following symbols have been used in this manual to help communicate the intent of the instructions. When one or more of the symbols appear at the beginning of a maintenance procedure, it conveys the meaning below.



Indicates that tools will be required to perform this procedure.



Indicates that new parts will be required to perform this procedure.

5.1.2 Pre-start inspection

- a) Be sure operator's, safety, and responsibilities manuals are complete, legible, and in the storage, container located on the machine.
- b) Be sure all decals are in place and legible.
- c) Check for hydraulic oil leaks and proper oil level. Add oil if needed. See Maintenance section.
- d) Check for battery fluid leaks and proper fluid level. Add distilled water if needed after battery charged. Check the following components or areas for damage, improperly installed, or missing parts and unauthorized modifications:

- 1) Electrical components, wiring, and electrical cables.
- 2) Hydraulic hoses, fittings, cylinders, and manifolds.
- 3) Drive motor/motor.
- 4) Wear pads.
- 5) Tires and wheels.
- 6) Limit switches and horn.
- 7) Alarm and indicator (if equipped).
- 8) Nuts, bolts and other fasteners.
- 9) Brake release unit.

5.1.3 Maintenance hazard

- a) Shut off power to all controls and ensure that all moving parts are secured from inadvertent motion prior to performing any adjustments or repairs.
- b) Never work under an elevated platform until it has been fully lowered to stowed position, if possible, or otherwise supported and restrained from movement with appropriate safety props, blocking, or overhead supports.
- c) DO NOT attempt to repair or tighten any hydraulic holes or fittings while the machine is powered on or when the hydraulic system is under pressure.
- d) Always relieve hydraulic pressure from all hydraulic circuits before loosening or removing hydraulic components.
- e) DO NOT use your hand to check for leaks. Use a piece of card- board or paper to search for leaks. Wear gloves to help protect hands from spraying fluid.



5.1.4 Body injury hazard

Do not operate a machine with a hydraulic oil or air leak. An air leak or hydraulic leak can penetrate and/or burn skin. During or after a period of running of the hydraulic system, the parts may produce high surface temperature, and improper contact will cause burn to skin. Overhauling or adjusting any part of hydraulic system can cause serious injuries. Only trained maintenance personnel are allowed to repair or adjust the hydraulic system.

Access by the operator is only advised when performing a pre-operation inspection. All compartments must remain closed and secured during operation.

5.2 Tire and Wheel

5.2.1 Tire replacement

ZOOMLION recommends a replacement tire be the same size, ply and brand as originally installed on the machine. Please refer to the ZOOMLION Parts Manual for the part number of the approved tires for a particular machine model. If not using a ZOOMLION approved replacement tire, we recommend that replacement tires have the following characteristics:

- a) Equal or greater ply/load rating and size of original;
- b) Tire tread contact width equal or greater than original;
- c) Wheel diameter, width, and offset dimensions equal to the original;
- d) Approved for the application by the tire manufacturer (including inflation pressure and maximum tire load).

Unless specifically approved by ZOOMLION, do not replace a foam filled tire assembly with a pneumatic tire. When selecting and installing a replacement tire, ensure that all tires are inflated to the pressure recommended by ZOOMLION. Due to size variations between tire brands, both tires on the same axle should be the same.

5.2.2 Wheel and tire requirements

The rims installed on each product model have been designed for stability requirements which consist of track width, tire pressure, and load capacity. Size changes such as rim width, center piece location, larger or smaller diameter, etc., without written factory recommendations, may result in an unsafe condition regarding stability.

5.2.3 Wheel installation

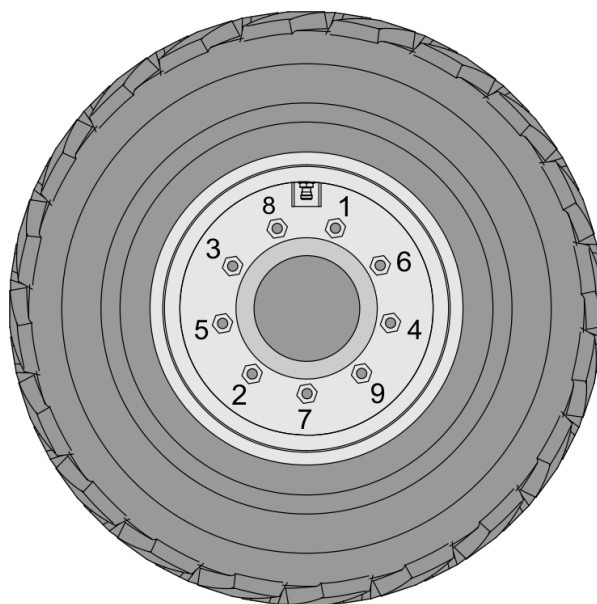
It is extremely important to apply and maintain proper wheel mounting torque.

WARNING

Wheel nuts must be installed and maintained at the proper torque to prevent loose wheels, broken studs, and possible dangerous separation of wheel from the axle. Be sure to use only the nuts matched to the cone angle of the wheel.

Tighten the lug nuts to the proper torque to prevent wheels from coming loose. Use a torque wrench to tighten the fasteners. If you do not have a torque wrench, tighten the fasteners with a lug wrench, then immediately have a service garage or dealer tighten the lug nuts to the proper torque. Over-tightening will result in breaking the studs or permanently deforming the mounting stud holes in the wheels. The proper procedure for attaching wheels is as follows:

- a) Start all nuts by hand to prevent cross threading. DO NOT use lubricant on threads or nuts.
- b) Tighten nuts in the following sequence.


Figure 5-20 Hub bolts

- c) The tightening of the nuts should be done in stages. Following the recommended sequence, tighten nuts per wheel torque.

Table 5-1 Wheel torque table

Tightening sequence		
First stage	Second stage	Third stage
130 Nm	230 Nm	400 Nm

- d) Wheel nuts should be torqued after first 50 hrs of operation and after each wheel removal. Check and tighten every 3 months or 150 hrs of operation.

5.3 Hydraulic System Maintenance



5.3.1 Hydraulic oil level

Maintaining the hydraulic oil at the proper level is essential to machine operation. Improper hydraulic oil levels can damage hydraulic components. Daily checks allow the inspector to identify changes in oil level that might indicate the presence of hydraulic system problems.

- a) Be sure the machine is on a firm and level surface and in stowed position.
- b) By observing oil level in hydraulic oil tank, the hydraulic oil level after excluding air in the hydraulic system should reach the maximum scale mark on the hydraulic oil tank, and not be higher than bottom of the oil tank cap (different models have different maximum scale).
- c) Add oil as needed. Do not overfill.

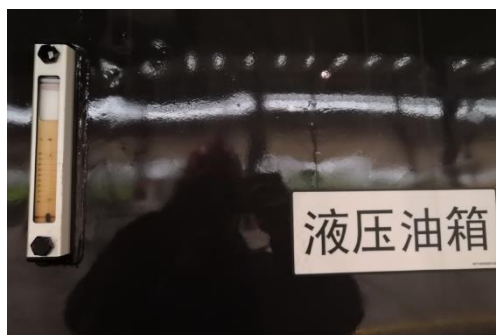


Figure 5-2 Hydraulic level gauge

5.3.2 Hydraulic oil capacity

Table 5-2 Capacity

Model	Hydraulic tank	Hydraulic system (including tank)
ZS1218ERT	44L	30L

5.3.3 Hydraulic oil specification

Please refer to the Table 5-4 for the recommended type and model of hydraulic oil (Table 5-7 is not used to specify the model and parameters of hydraulic oil). Please select proper hydraulic oil according to the specific application environment of the equipment. For special environment or special requirements of users, please contact ZOOMLION or the hydraulic oil manufacturer.

CAUTION

DO NOT mix oils of different brands or types, as they contain different additives which may cause negative effects. If mixing of hydraulic oils is unavoidable, permission must be obtained from the

hydraulic oil manufacturer. After-sales service of ZOOMLION does not cover machine malfunction caused by hydraulic oil mixing.

Table 5-3 Technical Parameters of Hydraulic Oil

Technical Parameters Type	ISO Viscosity Grade	Pour Point °C /°F	Flash Point °C /°F	kinematic viscosity cSt(40°C/10 4°F)	Viscosity index
Great Wall 4632 grease non- flammable hydraulic oil N32 (Eco-Friendly)	32	-20	270	28.8-35.2	180
Great Wall Ground NO.10 aviation hydraulic fluid	—	-55	107	10.53(50°C/ 122°F)	120
Great Wall Zhuoli L-HS 15 Ultralow temperature oil	15	-57	164	15.35	172
Great Wall Zhuoli L-HS 32 Ultralow temperature oil	32	-48	224	31.35	166
Great Wall Zhuoli L-HS 46 Ultralow temperature oil	46	-43	238	45.81	170
Great Wall Zhuoli L-HV 15 Low temperature oil	15	-45	173	15.51	140
Great Wall Zhuoli L-HV 32 Low temperature oil	32	-39	231	33.4	150
Great Wall Zhuoli L-HV 46 Low temperature oil	46	-37	240	48.7	150
Great Wall Zhuoli L-HV 68 Low temperature oil	68	-35	238	70.47	150
Great Wall Zhuoli L-HM 46 Antiwear hydraulic oil (High pressure)	46	15/5	240/464	45.8	97

Table 5-6 Technical parameter of hydraulic oil (continuous)

Technical Parameters Type	ISO Viscosity Grade	Pour Point °C /°F	Flash Point °C /°F	kinematic viscosity cSt(40°C/10 4°F)	Viscosity index
Great Wall Zhuoli L-HM 68 Antiwear hydraulic oil (High pressure)	68	13/9	245/473	67.4	98
Mobil SHC Aware H 32 (Eco- Friendly)	32	30/-22	185/365	32	140
Clarity Synthetic EA Hydraulic Oil	46	44/-47	221/430	46	180
Mobil DTE 10 Ultra 22	22	54/-65	224/435	22.4	164
Mobil DTE 10 Ultra 32	32	54/-65	250/482	32.7	164
Mobil DTE 10 Ultra 46	46	45/-49	232/450	45.6	164
Chevron/CaltexRando Rando HDZ 15	15	60/-76	150/302	15.7	144
Chevron/CaltexRando Rando HDZ 32	32	49/-56	204/399	33	150
Chevron/CaltexRando Rando HDZ 46	46	47/-53	216/421	46.7	153
Chevron/CaltexRando Rando MV 15	15	42/-44	154/309	15.8	155
Chevron/CaltexRando Rando MV 32	32	36/-33	210/410	33.5	154
Chevron/CaltexRando Rando MV 46	46	33/-27	214/417	44	154

5.3.4 Hydraulic Oil Viscosity and Temperature Limit

Proper use of hydraulic oil: please note the corresponding oil viscosity and temperature limit. Under normal conditions, the recommended oil temperature should be controlled at 30°C/86°F to 60°C/140°F, this highest temperature should be no more than 90°C/194°F. The oil temperature affects the oil viscosity and the thickness of the oil film. Excessive oil temperatures can also damage or shorten the life of the oil seal and other rubber components, which can cause oil leakage in the hydraulic system, while high temperatures can exacerbate the evaporation and oxidation of the oil.

Pre-delivery of the machine, specific model of hydraulic oil shall be added as required by customer. If machine operating environment temperature is beyond the temperature limit of the hydraulic oil, different hydraulic oil suited to the actual conditions shall be used in time. On account of the safety of machine components and work efficiency, it is advisable that the starting temperature should be 25°C/77°F higher than the pour point of hydraulic oil.

If the altitude is higher than 4,000 meters/13,100ft., please use a lower viscosity grade hydraulic oil based on the above requirements for the use of hydraulic oil for hydraulic pump normally operation.

5.3.5 Hydraulic oil replacement

We suggest that changing time of the hydraulic oil is as follows:

- a) First changing: operating for 500 hrs after commissioning.
- b) Second and subsequent changing: every 2,000 hrs of operation or once a year.

The above recommended intervals are suitable for most applications. Higher temperatures and pressures will shorten the oil's service life, so the hydraulic oil should be changed sooner than the recommended. For small load work, the oil change time can be extended.

Cleanliness of the hydraulic oil upon delivery is NAS9 (ISO4406 18/15), and for normal operation, the cleanliness should not be lower than NAS10 (ISO4406 19/16). We suggest that the hydraulic oil should be checked every 6 months, and the oil should be sampled at least once upon the time for oil changing. The oil sample can be sent to the hydraulic oil manufacturer or qualified third-party testing agency for analysis and to determine whether it is still usable.

5.3.6 Return oil filter element replacement

The return oil filter element is recommended to be replaced every 1000 hours or once a year, whichever comes first. A good working return filter element is very important to the performance and service life of the machine. Dirty or blocked filters may affect the performance of the machine, if continued use will lead to damage to the parts. In harsh environment or working conditions, should increase the filter inspection and filter cartridge replacement frequency.

5.4 Battery Maintenance



Proper battery condition is essential to good machine performance and operational safety. Improper fluid levels or damaged cables and connections can result in component damage and hazardous conditions.

This inspection is not required for machines with sealed or non-maintainable batteries.

Check electrolyte level of the battery every two weeks. Fully charge the battery before adding water. If the electrolyte level is much higher than the plate, then no need to add water.

⚠ CAUTION

Electric shock hazard:

Contact with hot or live circuits may result in death or serious injury. Remove all rings, watches and jewelry.

⚠ CAUTION

Body Injury Hazard:

Batteries contain acid. Avoid spilling or contacting battery acid. Neutralize battery acid spills with baking soda and water.

The battery should be fully charged before this inspection.

- a) Only qualified riggers should rig the machine.
- b) Only certified crane operators should lift the machine and only in accordance with the applicable crane regulations.
- c) Be sure that the battery hold-down brackets are in place and secure.

Adding terminal protectors and a corrosion preventative sealant will help eliminate the corrosion on the battery terminals and cables.

5.5 Regular Maintenance

- a) Maintenance performed quarterly, annually and every two years must be completed by a person trained and qualified to perform maintenance on this machine according to the procedures found in the service manual for this machine.
- b) Machines that have been out of service for more than three months must receive the quarterly inspection before they are put back into service.
- c) By observing and detecting the oil level in the hydraulic tank, after removing the air in the hydraulic system, the hydraulic oil level should reach the maximum scale mark on the hydraulic tank, not higher than the bottom end face of the tank cap (the maximum scale is different for different models);

- d) Fill hydraulic oil according to the actual needs of the hydraulic system, do not overfill;
- e) Torque check of tire mounting bolts every quarter.

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**Section 6 Storage and
Ex-factory Test**



SECTION 6 STORAGE AND EX-FACTORY TEST

6.1 Storage Conditions

Ambient temperature for machine storage and transportation should be between -20°C/-4°F and 40°C/104°F, with relative humidity not greater than 85% and 100% only for short-term.

6.2 Ex-factory Test Items

Machine must complete testing items in the following table before delivery:

Table 6-1 Ex-factory test item (ZS1218ERT)

Tests Items	Load Testing		Testing Movement
Overload Test	125%	437.5kg/965 lbs	Platform Lifting
Functional Test	110%	385kg/849 lbs	Traveling & Platform Lifting
Braking Test	100%	350kg/772 lbs	Max Speed of Forward & Reverse Traveling

ZOOMLION

Operation and Safety Manual

Section 7 Technical Parameter



SECTION 7 TECHNICAL PARAMETER

Table 7-1 ZS1218ERT Series Technical parameters

Model	ZS1218ERT	Parameters	
Dimension	Work Height	13.8 m	45ft 3in
	Platform Height	11.8 m	38ft 9in
	Max Height (stowed position)	2.74 m	9ft
	Max Platform Height (stowed position)	1.65 m	5ft 5in
	Max Height (Stowed, rails folded)	2.05 m	6ft 9in
	Rails Height	1.1 m	3ft 7in
	Overall Width	1.75 m	5ft 9in
	Extension Length	3.5 m	11ft 6in
	Extension Length	1.2 m	3ft 11in
	Platform Size (Length ×Width)	2.75 × 1.52 m	9ft×5ft
	Wheelbase	1.95 m	6ft 5in
	Ground Clearance	0.24 m	9.5in
Working Performance	Platform load rating Capacity	350 kg	770lb
	Max. Number of Workers	3 (Indoor) /2(Outdoor)	3 (Indoor) /2(Outdoor)
	Drive Speed (Stowed)	6 km/h	3.7mph
	Drive Speed (Elevated)	0.5 km/h	0.3mph
	Maximum slope rating (stowed)	40% (22°)	40% (22°)
	Maximum Side Slope Rating in Stowed	40% (22°)	40% (22°)
	Maximum chassis working Angle	3°/2°	3°/2°
	Turning Radius (Outside)	4.2 m	13ft 9in
	Turning Radius (Inside)	2.0 m	6ft 7in
	Total vibration value of the hand/arm system	≤2.5 m/s	≤5.6 mph
	Maximum square root value of the weighted	≤0.5 m/s ²	≤1ft 8in /s ²

Table 7-1 ZS1218ERT Series Technical parameters (cont.)

No.	ZS1218ERT	Parameters	
Working Condition	Max wind speed	12.5 m/s	28mph
	Lowest working temperature	-20°C	-4°F
	Highest working temperature	60°C	140°F
	Sound pressure level on ground	<70 dBA	<70 dBA
	Sound pressure level on platform	<70 dBA	<70 dBA
Other	Machine weight	4900kg	10800lbs
	Power	8 packs of lithium batteries, 6V/230AH	8 packs of lithium batteries, 6V/230AH
	System voltage	48V	48V
	Platform AC socket	Standard	Standard
	Max hydraulic pressure	290 bar	4210psi□
	Tire size	27×12-15/8.0 (solid)	27×12-15/8.0 (solid)
	Max tire load	1350 kg	2980lbs
	Max outrigger load	2460kg	5420lbs

Appendix: Inspection and Maintenance Records

Date	record

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Date	record

Electric Rough-Terrain Scissor Lifts

Operation and Safety Manual

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